M.A.M. COLLEGE OF ENGINEERING AND TECHNOLOGY

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CS3811 Project Work

PROJECT TITLE : QR-AUTHENTICATED BLOCKCHAIN ARCHITECTURE FOR PROTECTED GENOMIC AND MEDICAL RESEARCH DATA SHARING

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MODEL REVIEW

PROJECT TITLE : QR-AUTHENTICATED BLOCKCHAIN ARCHITECTURE FOR PROTECTED GENOMIC AND MEDICAL RESEARCH DATA SHARING

CONTENT IN THIS PRESENTATION

PROBLEM DEFINITION

- Genomic and medical research data are highly sensitive and prone to unauthorized access, misuse, and privacy breaches.
- During transmission across multiple networks, such data is vulnerable to hacking, cyber-attacks, and interception.
- Traditional centralized storage systems cannot guarantee tamper-proof or immutable records, risking data integrity.
- Existing systems lack a robust approval and access control mechanism, making it difficult to restrict data access to authorized buyers only.
- There is a need for a secure, transparent, and traceable platform to share sensitive genomic data safely between owners and buyers.

OBJECTIVE

- To securely store sensitive genomic and medical research data using encryption and blockchain technology.
- To implement a QR-based authentication system for verifying the identity of data owners and buyers.
- To provide a controlled access mechanism where owners can approve or reject buyers’ requests.
- To ensure that all transactions, including uploads, approvals, downloads, and key exchanges, are tamper-proof and traceable via blockchain.
- To create a reliable and transparent platform for safe genomic data sharing that prevents unauthorized access and data breaches.

LITERATURE SURVEY

Sl. No.	Details of the paper	
1	Data protection and privacy of the internet of healthcare things (IoHTs)	
Objective	Methodology	Result
Analyze data protection and privacy challenges in IoHT systems.	Reviewed IoHT architectures and security mechanisms.	Highlights the importance of secure data handling and privacy preservation in healthcare IoT environments.

LITERATURE SURVEY

Sl. No.	Details of the paper	
2	Ethics of artificial intelligence in education: Student privacy and data protection	
Objective	Methodology	Result
Examine ethical issues and privacy concerns of AI applications in education.	Analytical review of AI tools in educational settings focusing on student data.	Emphasizes the need for strict privacy measures when handling sensitive student data.

LITERATURE SURVEY

Sl. No.	Details of the paper	
3	Data privacy laws and compliance: a comparative review of the EU GDPR and USA regulations	
Objective	Methodology	Result
Compare data privacy laws and compliance requirements between the EU and USA.	Comparative analysis of GDPR and US privacy regulations.	Provides insights into global data protection standards, useful for designing compliant systems.

LITERATURE SURVEY

Sl. No.	Details of the paper	
4	Wind power forecasting considering data privacy protection: A federated deep reinforcement learning approach	
Objective	Methodology	Result
Forecast wind power while maintaining data privacy.	Implemented federated deep reinforcement learning to protect sensitive data.	Demonstrates the effectiveness of federated learning in secure data handling across distributed networks.

LITERATURE SURVEY

Sl. No.	Details of the paper	
5	Toward a comprehensive framework for ensuring security and privacy in artificial intelligence	
Objective	Methodology	Result
Propose a framework to secure AI systems and protect user privacy.	Developed a security and privacy framework combining multiple AI defense mechanisms.	Highlights strategies for maintaining AI system integrity and sensitive data protection.

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

- **Processor:** Intel i5 or higher / AMD equivalent
- **RAM:** Minimum 8GB (16GB recommended)
- **Storage:** SSD with at least 256GB (512GB recommended)
- **Graphics:** Integrated GPU (Dedicated GPU for AI-related tasks)
- **Internet Connection:** Stable broadband for cloud-based functionalities

SOFTWARE REQUIREMENTS

- **Operating System:** Windows, macOS, or Linux
- **Programming Languages:** JavaScript, Python
- **Frontend Technologies:** React.js, Tailwind CSS
- **Backend Technologies:** Node.js, Express.js
- **Database:** Supabase
- **AI & Automation Tools:** OpenAI API, Inngest
- **Development Tools:** VS Code, Git, GitHub
- **Security Tools:** Arcjet

FEASIBILITY ANALYSIS

- The system leverages well-established technologies like blockchain, AES/ECC encryption, and QR-based authentication, making it technically feasible.
- Open-source blockchain platforms, encryption libraries, and QR tools make the project economically viable with minimal software costs.
- The modular design ensures smooth integration of registration, encryption, storage, access control, and decryption functionalities, supporting operational feasibility.
- By securing sensitive genomic data and controlling access, the system reduces potential financial losses from hacking or unauthorized use, increasing economic feasibility.
- Current software frameworks and standard hardware requirements allow the project to be implemented within a reasonable timeframe, ensuring practical deployment feasibility.

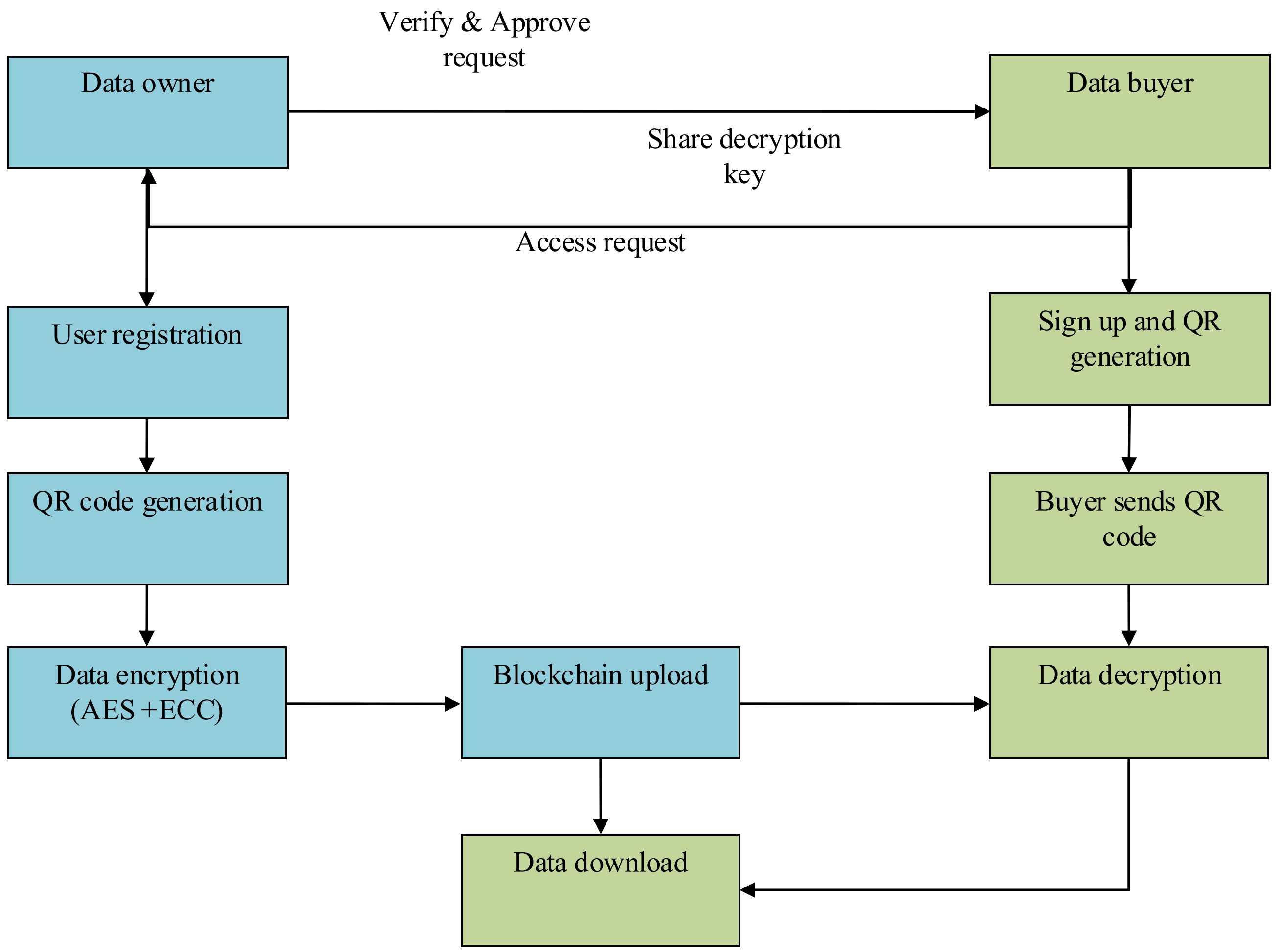
EXISTING SYSTEM

- Current genomic and medical data sharing relies mostly on centralized storage systems, which are vulnerable to hacking and data breaches.
- Data is often transmitted over insecure networks without strong encryption, increasing the risk of interception and unauthorized access.
- Existing systems usually lack role-based access control, making it difficult to restrict sensitive data to authorized buyers only.
- There is no permanent audit trail for data transactions, making it hard to track downloads, approvals, or modifications.
- Traditional methods provide limited transparency and traceability, reducing trust among owners, buyers, and stakeholders.

PROPOSED SYSTEM

- The system securely stores sensitive genomic and medical research data using encryption with hybrid algorithm (AES + ECC) before uploading it to a blockchain-based decentralized storage.
- Genomic data owners can register, upload encrypted files, and manage access permissions for authorized buyers.
- Buyers register in the system, receive a unique QR code, and can request access to specific genomic datasets.
- Owners approve buyers' requests, and the system provides them with a decryption key to securely access and download the data.
- All transactions, including uploads, approvals, downloads, and key exchanges, are permanently recorded in the blockchain, ensuring traceability, transparency, and tamper-proof access.

BLOCK DIAGRAM



MODULE DESCRIPTION

- User Registration and QR Generation Module
- Data Encryption and Upload Module
- Blockchain Storage Module
- Buyer Approval and Access Control Module
- QR-Based Authentication Module
- Data Decryption and Download Module

MODULE DESCRIPTION

1. User Registration and QR Generation Module

- Allows genomic data owners and buyers to create an account in the system.
- Collects necessary user information like name, email, organization, and role (owner or buyer).
- Generates a unique QR code for each user to serve as a secure identity verification token.
- Stores user registration data securely in the database.
- Prepares users for secure login and access to the blockchain-based data system.

MODULE DESCRIPTION

2. Data Encryption and Upload Module

- Enables owners to select genomic or medical research data for sharing.
- Encrypts the selected data using secure encryption algorithms (e.g., AES or ECC).
- Ensures that the data is transformed into a secure, unreadable format before uploading.
- Prepares the encrypted data for blockchain storage by generating a transaction hash.
- Provides a secure interface to upload the encrypted data into the blockchain ledger.

MODULE DESCRIPTION

3. Blockchain Storage Module

- Stores all encrypted genomic data in a decentralized blockchain ledger.
- Records transaction details such as upload time, data hash, and owner identity.
- Ensures that the stored data is tamper-proof and immutable.
- Maintains transparency and traceability for all data transactions.
- Supports secure retrieval of data for authorized buyers after approval.

MODULE DESCRIPTION

4. Buyer Approval and Access Control Module

- Receives data access requests from registered buyers.
- Enables owners to approve or reject each buyer's request.
- Controls which buyers can access specific genomic data.
- Sends notifications to buyers upon approval or rejection.
- Integrates with the blockchain to record all approval transactions for auditing.

MODULE DESCRIPTION

5. QR-Based Authentication Module

- Allows owners and buyers to log in securely using their unique QR codes.
- Validates the identity of each user before granting system access.
- Prevents unauthorized access or impersonation within the platform.
- Works in combination with blockchain to verify the legitimacy of each transaction.
- Ensures that only approved users can request or download encrypted genomic data.

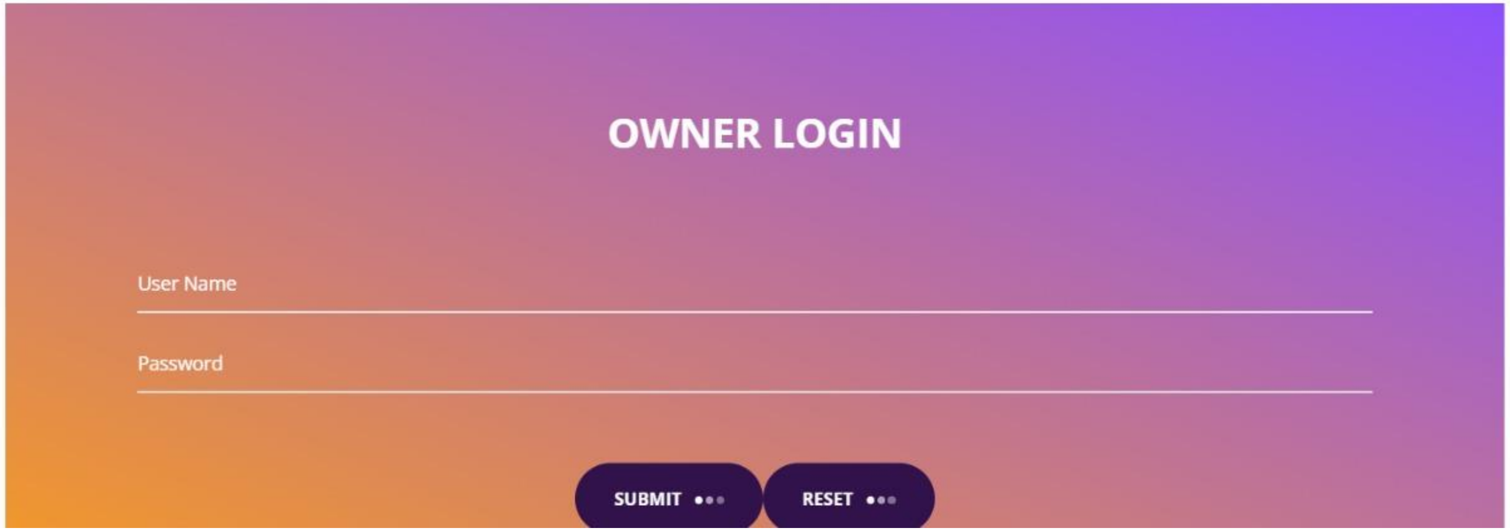
MODULE DESCRIPTION

6. Data Decryption and Download Module

- Provides approved buyers with a secure decryption key after owner approval.
- Enables buyers to decrypt the downloaded encrypted genomic data safely.
- Ensures that data remains confidential during download and decryption.
- Supports logging of download and decryption activities in the blockchain for traceability.
- Allows buyers to access and use the genomic data for authorized research purposes only.

SCREENSHOTS

SCREENSHOTS



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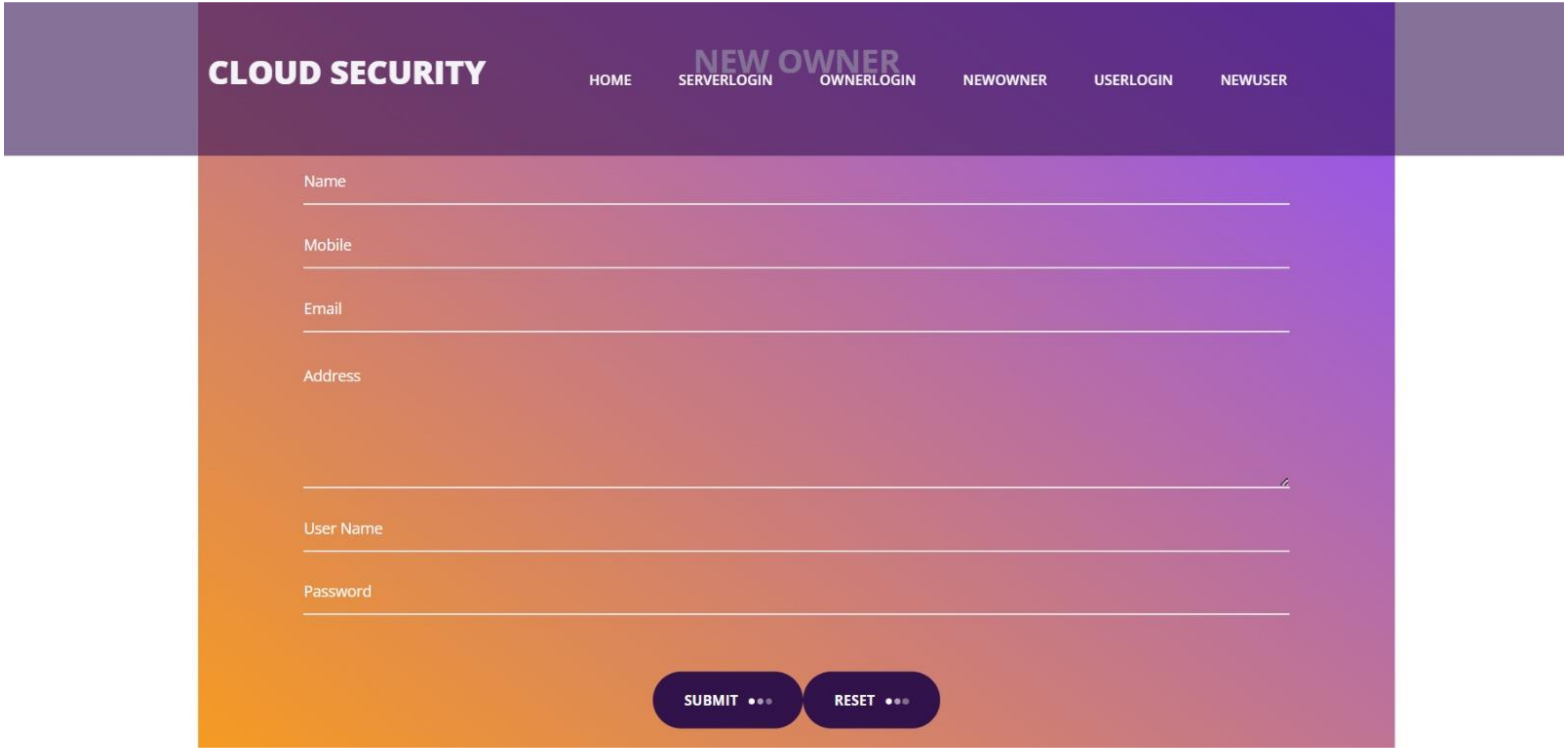

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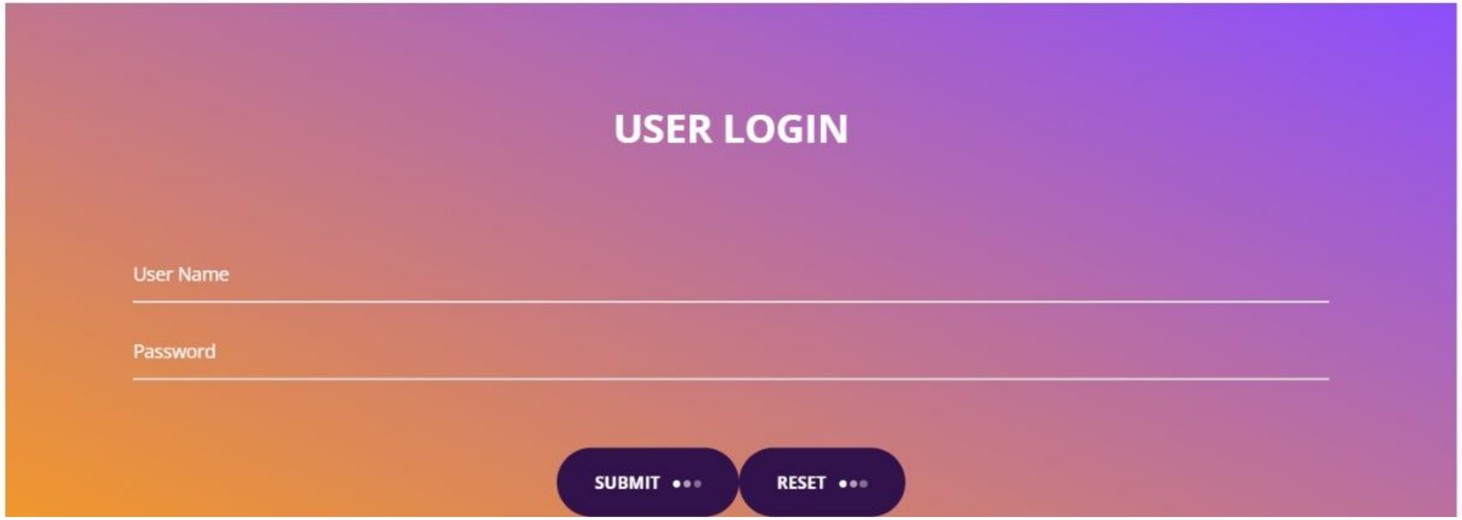

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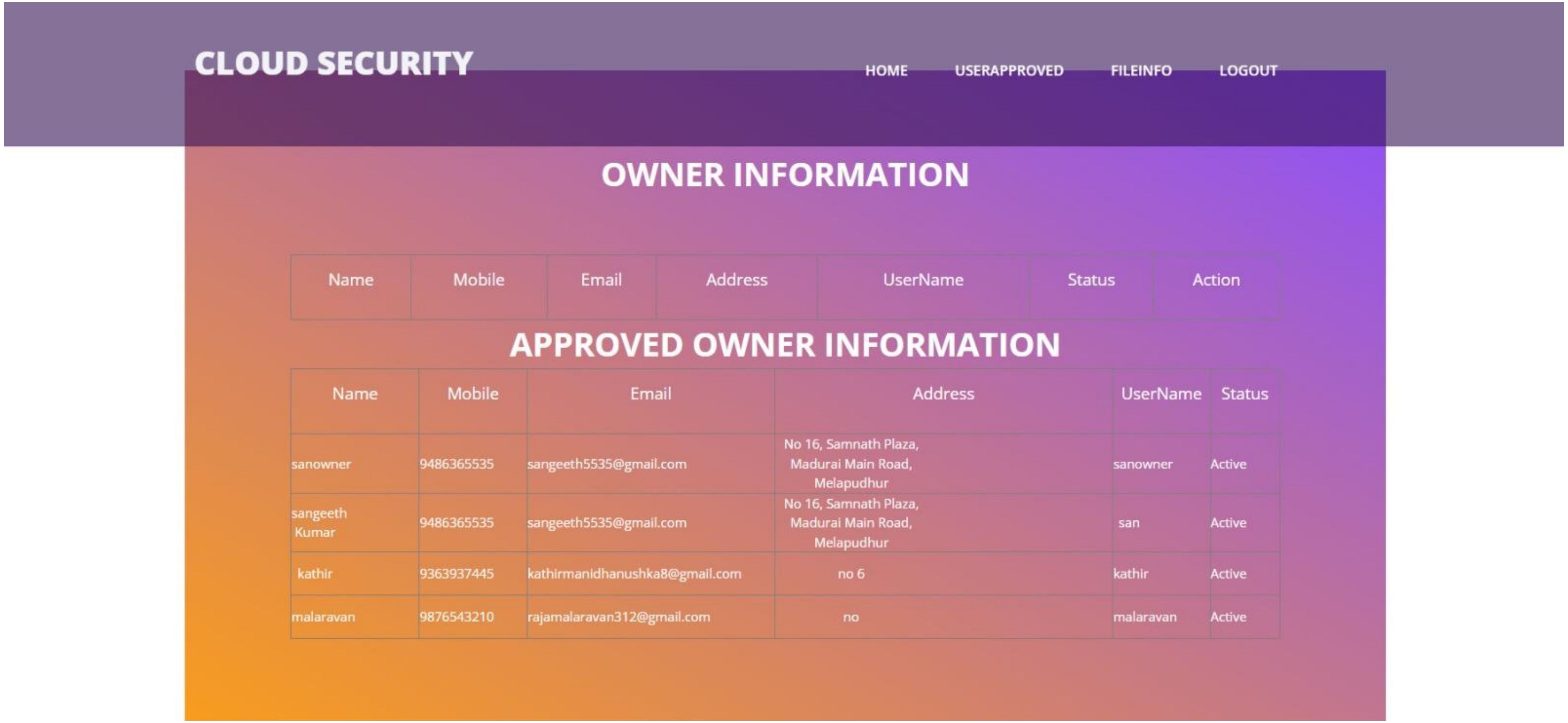

CLOUD SECURITY

HOME USERAPPROVED FILEINFO LOGOUT

OWNER INFORMATION

Name	Mobile	Email	Address	UserName	Status	Action

APPROVED OWNER INFORMATION

Name	Mobile	Email	Address	UserName	Status
sanowner	9486365535	sangeeth5535@gmail.com	No 16, Samnath Plaza, Madurai Main Road, Melapudhur	sanowner	Active
sangeeth Kumar	9486365535	sangeeth5535@gmail.com	No 16, Samnath Plaza, Madurai Main Road, Melapudhur	san	Active
kathir	9363937445	kathirmanidhanushka8@gmail.com	no 6	kathir	Active
malaravan	9876543210	rajamalaravan312@gmail.com	no	malaravan	Active

FUTURE ENHANCEMENT

- Integration of advanced biometric authentication (e.g., face recognition or fingerprint) alongside QR codes for stronger user verification.
- Implementation of AI-based anomaly detection to identify unauthorized access attempts or suspicious activities in real-time.
- Expansion to support large-scale genomic databases with high-speed blockchain networks for faster data transactions.
- Development of a mobile application for seamless access, request approvals, and secure downloads on smart phones or tablets.
- Incorporation of smart contracts to automate buyer approvals, data access, and key distribution securely without manual intervention.

CONCLUSION

- The proposed system provides a secure and tamper-proof platform for sharing sensitive genomic and medical research data.
- Integration of encryption and blockchain ensures data confidentiality, integrity, and transparency during storage and transmission.
- QR-based authentication and access control effectively verify users and restrict data access to authorized buyers only.
- All transactions, including uploads, approvals, downloads, and key exchanges, are permanently recorded in the blockchain for traceability.
- The system enhances trust, reduces the risk of data breaches, and offers a reliable framework for safe genomic data sharing among owners and buyers.

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THANK YOU